Hypothesis Testing (Part 1)

STAT 4530/6530
Statistical Inference for Data Scientists

Hypothesis Testing

In statistics, a **hypothesis** is a statement about a population distribution.

It takes the form of a prediction about the distribution, such as that a particular parameter has the same value for two groups being compared.

Examples:

- For all the employees of a fast-food chain, the mean salary is the same for women and for men
- For all college students in the U.S., there is no correlation between weekly number of hours spend studying and the number of hours spent partying
- Only half of adults in the United Kingdom are satisfied with the National Health Service

The hypotheses are investigated with reference to alternative hypotheses, such as "For all employees of a fast-food chain, the mean salary is higher for men than for women."

Example: Testing for Bias in Selecting Managers

A statistical **significance test** (often simply referred to as a "test") uses data to summarize the evidence about a hypothesis by comparing a point estimate of the parameter of interest to the value predicted by the hypothesis.

Consider the following example:

- A supermarket chain in the southeastern U.S. periodically selects employees to receive management training.
- A group of female employees recently asserted that the company selects males at a disproportionately high rate for such training.
- The company denied the claim.

Example: Testing for Bias in Selecting Managers

Suppose the employee pool for potential management training is large and has 40% males and 60% females.

- The company's claim of a lack of gender bias is a hypothesis test that, all else being equal, at each choice the probability of selecting a male equals 0.40 and the probability of selecting a female equals 0.60.
- The claim of the aforementioned group of female employees is an alternative hypothesis that the probability of selecting a male exceeds 0.40.
- Suppose that in the latest selection of employees for management training, 9 out of 10 chosen were male.

Example: Testing for Bias in Selecting Managers

Suppose that in the latest selection of employees for management training, 9 out of 10 chosen were male.

- 0.90 is a much higher probability than the expected 0.40. On the other hand, even if selected at random from the employees pool, due to sampling variation, not exactly 40% of those chosen would necessarily end up being male. But would ending up with 90% chosen being male be plausible?
- We want to analyze how unlikely it is for 9 out of the 10 selected employees to be male if there were *no* gender bias.

Significance Tests: Assumptions

Each significance test has four elements: (1) Assumptions, (2) Hypotheses, (3) Test statistic, and (4) P-value and conclusion.

Assumptions: each significance test makes certain assumptions that must hold in order for it to be valid. These pertain to:

- **Type of data:** are the data quantitative or categorical?
- **Randomization:** each significance test assumes that the data gathering employed randomization, such as a simple random sample or a randomized experiment.
- **Population distribution:** some significance tests assume that the variable has population distribution in a particular family of probability distributions (e.g., normal or Bernoulli)
- **Sample size:** some significance tests, such as those based on the central limit theorem, are valid only when the sample size is sufficiently large.

Significance Tests: Hypotheses

Hypotheses: significance tests have two hypotheses about population distributions, usually expressed in terms of parameters of those distributions.

- **Null hypothesis:** denoted by H_0, is a statement that a parameter takes a particular value.
- **Alternative hypothesis:** denoted by H_a, states that a parameter takes a value in some alternative range.
- Usually, the value in H_0 corresponds, in a certain sense, to *no effect*. The values in H_a then represent a particular type of effect.
- The hypotheses are formulated *before* analyzing the data. The alternative hypothesis is usually a research hypothesis that the investigator believes to be true.

Example: Testing for Bias in Selecting Managers

Let us return to the example of testing for bias among selecting managers.

- Let p denote the probability that any particular selection is male.
- The company's claim that $p = 0.40$ is an example of a null hypothesis: no gender bias effect.
- The alternative hypothesis reflects the female employees' belief that this probability exceeds 0.40.
- The hypotheses are denoted $H_0 : p = 0.40$ and $H_a : p > 0.40$. Whereas H_0 has a single value, H_a has a range of values.

Example: Testing for Bias in Selecting Managers

A significance test analyzes whether the data contradict H_0 and suggest H_a is true using *proof by contradiction*.

- H_0 is presumed to be true. Under this assumption, if the observed data would be very unusual, the evidence would contradict H_0 and support H_a.
- In testing potential gender bias, we presume that $H_0 : p = 0.40$ is true and analyze whether a sample proportion of males of 0.90 would be unusual under this presumption. If it is, we may be inclined to reject H_0 and believe H_a.
- If, on the other hand, the difference between the sample proportion 0.90 and the H_0 value of 0.40 could easily be due to ordinary sampling variability, we conclude that the company's claim of a lack of gender bias is plausible.

Significance Tests: Test Statistic

The parameter to which the hypothesis refer has a point estimate.

- The **test statistic** summarizes how far the point estimate of the parameter falls from the parameter value specified by H_0.
- The distance between the H_0 parameter value and point estimate of the parameter can be quantified by the number of standard errors between them.

Significance Tests: *P*-value and Conclusion

We want a probabilistic summary of the evidence against H_0.

- We can use the sampling distribution of the test statistic, under the presumption that H_0 is true.
- The farther the test statistic falls out in a tail of the sampling distribution in the direction predicted by H_a, the stronger the evidence against H_0.
- The **P-value** is the probability, presuming that H_0 is true, that the test statistic equals the observed value or a value even more extreme in the direction predicted by H_a.
- A small P-value, such as 0.01, means that the observed data would have been unusual if H_0 were actually true. A moderate to large P-value, such as 0.26 or 0.83, means that the data are consistent with H_0: if H_0 were in fact true, the observed data would not be unusual.
- In summary, smaller P-values reflect stronger evidence against H_0.

Example: Testing for Bias in Selecting Managers

For testing $H_0 : p = 0.40$ against $H_a : p > 0.40$ for the managerial trainee selections, we can use the sample count $y = 9$ of males chosen (out of a total of $n = 10$ selections) as the test statistic.

The values of y that provide this much or more extreme evidence against H_0 are the right-tail values greater than or equal to 9. From the binomial distribution, the probability of selecting 9 or 10 males in 10 selections when H_0 is true is:

$$P(Y \geq 9) = \binom{10}{9}(0.4)^9(0.6)^1 + \binom{10}{10}(0.4)^{10}(0.6)^0 = 0.0017.$$

Thus, the P-value is 0.0017, meaning that if the selections were truly random with respect to gender, the probability of such an extreme sample result is only 0.0017. This provides strong evidence against $H_0 : p = 0.40$ and supporting $H_a : p > 0.40$.

Significance Tests: P-value and Conclusion

Sometimes it is necessary to make a decision about the validity of H_0.

- If the P-value is sufficiently small, we reject H_0 and accept H_a.
- Research studies require small P-values, such as ≤ 0.05, in order to reject H_0.
- Results are then said to be *significant at the 0.05 level*.

The Elements of a Significance Test for a Proportion

Assumptions: We assume the data are obtained using randomization. The sample size should be sufficiently large that the sampling distribution of the sample proportion $\overline{y}$ is approximately normal. This is satisfied when the expected number of observations is at least 10 for both categories.

- For example, to test $H_0 : p = 0.50$, we need at least about $n = 20$.
- To test $H_0 : p = 0.90$ or $H_0 : p = 0.10$, we need at least $n = 100$.
- With small n or for greater precision, we can use the binomial distribution directly instead of a normal sampling distribution (as we did in our gender-bias example).

The Elements of a Significance Test for a Proportion

Hypotheses: The null hypothesis of a test about a population has the form $H_0 : p = p_0$.

- Here, p_0 denotes a particular value between 0 and 1
- The most common alternative hypothesis is of the form $H_a : p \neq p_0$. Such an alternative hypothesis is called **two-sided** because it contains values both below and above the value specified in H_0.

The Elements of a Significance Test for a Proportion

Test Statistic: The sampling distribution of the sample proportion $\overline{y}$ is approximately normal with mean p and standard error $\sqrt{p(1-p)/n}$.

- When $H_0 : p = p_0$ is true, the standard error is $se_0 = \sqrt{p_0(1-p_0)/n}$
- The evidence about H_0 is summarized by the number of standard errors that $\overline{y}$ falls from p_0.
- The test statistic is
$$Z = \frac{\overline{y} - p_0}{se_0}.$$
- When H_0 is true, its sampling distribution is approximately the standard normal distribution

The Elements of a Significance Test for a Proportion

P**-value and Conclusion:** Under the presumption that H_0 is true, the P-value is the probability that Z equals the observed value or a more extreme value that provides even stronger evidence against H_0 in favor of H_a.

- For $H_a : p \neq p_0$, the more extreme z values are the ones even farther out in the two tails of the standard normal distribution, and the P-value is the two-tail probability that $|Z|$ is at least as large as the observed $|z|$.
- This is also the probability that $\overline{y}$ falls at least as far from p_0 in either direction as the observed value of $\overline{y}$.

Reference

Agresti, A., & Kateri, M. (2021). Foundations of Statistics for Data Scientists: With R and Python (1st ed.). Chapman and Hall/CRC.